# Milestone 3 | E05

## Unity Catalog Email Column Mask

SUBMISSION TEMPLATE

| Difficulty | Marks | Tasks | Working budget |
|---|---|---|---|
| Easy | 10 | 4 | 25 min |

Trainee name: _____Soumiya Ayyappan_____ Candidate ID: _____10877547_____

Assigned catalog: _____ Date: _____

Workspace / pipeline / dashboard location: _____

### Business objective

Create an email masking policy that preserves NULLs and verifies masked read results.

### Screenshot and evaluator evidence policy

Screenshots are NOT limited to one per task. Include as many clear screenshots as necessary to prove every rubric criterion. If the evaluator checks CODE and RESULT, submit BOTH separately; code-only or output-only evidence is insufficient. Capture sequential code parts (A/B/C) if lengthy, without shrinking text. UI/metadata screenshots must reveal selected object identity, settings, or execution status. Use readable screenshots, not stitched micro-text. Do not rerun mutations to capture evidence. No notebook/source-file attachment replaces screenshots.

### Task requirements and evidence holders

### Task 1 - Prepare source volume

Marks: 2 | Evidence: UI

Required work: Create question-specific external volume and upload input CSV under input.

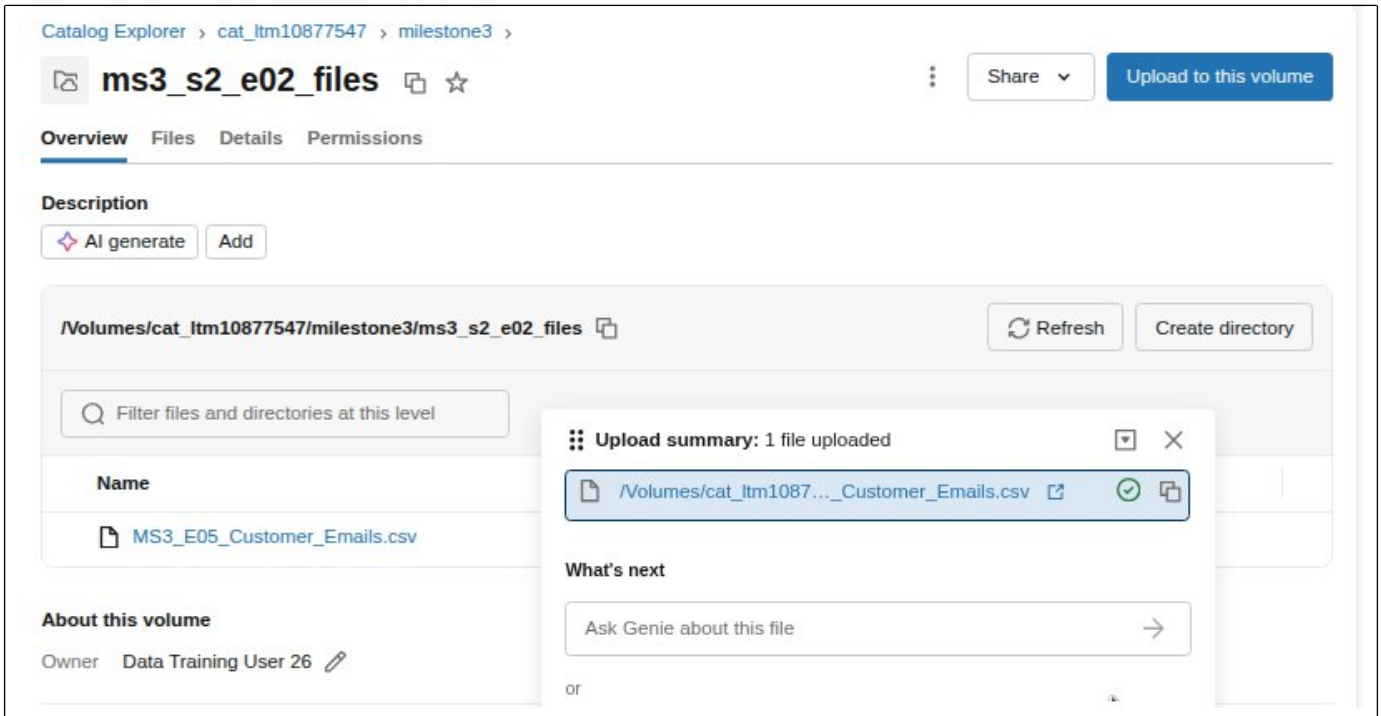
Expected result: 6 rows and correct source filename at the assigned volume location.

Evaluator checks: Correct governed source setup.

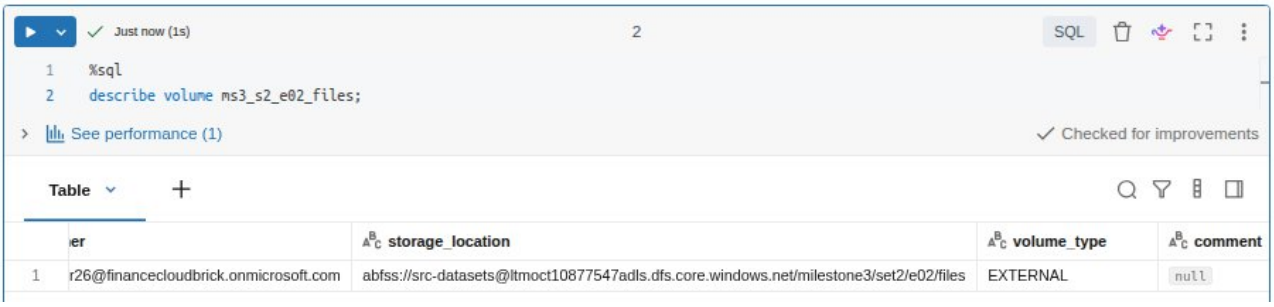
Screenshot instructions: Use MS3_S2_E02_T01_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 1 BELOW:

| Screenshot holder (paste image here; duplicate this holder if you need multiple images) |
|---|
| |


Catalog Explorer › cat_ltm10877547 › milestone3 ›
ms3_s2_e02_files
Share
Upload to this volume
Overview
Files
Details
Permissions
Description
AI generate
Add
/Volumes/cat_ltm10877547/milestone3/ms3_s2_e02_files
Refresh
Create directory
Filter files and directories at this level
Upload summary: 1 file uploaded
/Volumes/cat_ltm1087..._Customer_Emails.csv
Name
MS3_E05_Customer_Emails.csv
What's next
Ask Genie about this file
or
About this volume
Owner
Data Training User 26


current_catalog()
current_schema()
1
cat_ltm10877547
milestone3
1 row
9.070s runtime
Refreshed at 10/29/26, 11:39 AM (Now)
This result is stored as _sqldf and can be used in other Python and SQL cells.


Just now (1s)
2
SQL
1 %sql
2 describe volume ms3_s2_e02_files;
See performance (1)
Checked for improvements
Table
storage_location
volume_type
comment
1
r26@financecloudbrick.onmicrosoft.com
abfss://src-datasets@ltmoct10877547adls.dfs.core.windows.net/milestone3/set2/e02/files
EXTERNAL
null

```python
from pyspark.sql.functions import *
from pyspark.sql.types import *
schema = StructType([
    StructField("customer_id", StringType(), True),
    StructField("customer_name", StringType(), True),
    StructField("email", StringType(), True, metadata={"nullable": True}),
    StructField("region", StringType(), True),
])
path="/Volumes/cat_ltm10877547/milestone3/ms3_s2_e02_files/MS3_E05_Customer_Emails.csv"
df=spark.read.csv(path,header=True,schema=schema)
display(df.count())
display(df)
```

```
df:  pyspark.sql.connect.dataframe.DataFrame = [customer_id: string, customer_name: string ... 2 more fields]
6
```

Table +

| | customer_id | customer_name | email | region |
|---|---|---|---|---|
| 1 | CE01 | Anita | anita@example.com | NORTH |
| 2 | CE02 | Bala | bala@example.com | SOUTH |
| 3 | CE03 | Charu | null | NORTH |
| 4 | CE04 | Deepak | deepak@example.c... | WEST |
| 5 | CE05 | Esha | esha@example.com | SOUTH |
| 6 | CE06 | Farhan | null | WEST |

6 rows | 1.259s runtime

Evidence filename(s) / workspace reference: _____

### Task 2 - Create source Delta table

Marks: 3 | Evidence: Code + Result

Required work: Load four typed columns into a managed Delta table, keeping blank email cells as NULL rather than empty text.

Expected result: 6 rows; CE03 and CE06 have NULL email.

Evaluator checks: Correct managed table and NULL handling.

Screenshot instructions: Use MS3_S2_E02_T02_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 2 BELOW:

Screenshot holder (paste image here; duplicate this holder if you need multiple images)

```python
from pyspark.sql.functions import col, when, trim

path = "/Volumes/cat_ltm10877547/milestone3/ms3_s2_e02_files/MS3_E05_Customer_Emails.csv"

schema = "customer_id STRING, customer_name STRING, email STRING, region STRING"
df = (
    spark.read.csv(path, header=True, schema=schema)
    .withColumn("email", when((col("email").isNull()) | (trim(col("email")) == ""), None).otherwise(col("email")))
)
df.write.format("delta").mode("overwrite").saveAsTable("cat_ltm10877547.milestone3.ms3_s2_e05_customer_emails")
result = spark.table("cat_ltm10877547.milestone3.ms3_s2_e05_customer_emails")
print(f"Row count: {result.count()}")
display(result.orderBy("customer_id"))
```

```
Row count: 6
```

Table +

| | customer_id | customer_name | email | region |
|---|---|---|---|---|
| 1 | CE01 | Anita | anita@example.com | NORTH |
| 2 | CE02 | Bala | bala@example.com | SOUTH |
| 3 | CE03 | Charu | null | NORTH |
| 4 | CE04 | Deepak | deepak@example.c… | WEST |
| 5 | CE05 | Esha | esha@example.com | SOUTH |
| 6 | CE06 | Farhan | null | WEST |

Evidence filename(s) / workspace reference: _____

### Task 3 - Create and attach column mask

Marks: 3 | Evidence: Code + UI

Required work: Create a scalar Unity Catalog SQL masking function with STRING input/output and attach it directly to email; retain underlying data.

Expected result: Column metadata shows mask attached to email; masking function handles NULL.

Evaluator checks: Correct function and column-mask attachment.

Screenshot instructions: Use MS3_S2_E02_T03_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 3 BELOW:

| Screenshot holder (paste image here; duplicate this holder if you need multiple images) |
|---|

```
spark.sql("""
CREATE OR REPLACE FUNCTION cat_ltm10877547.milestone3.mask_email(email_val STRING)
RETURNS STRING
RETURN
  CASE
    WHEN email_val IS NULL THEN NULL
    ELSE CONCAT(LEFT(email_val, 1), '****@', SUBSTRING_INDEX(email_val, '@', -1))
  END
""")

spark.sql("""
ALTER TABLE cat_ltm10877547.milestone3.ms3_s2_e05_customer_emails
ALTER COLUMN email SET MASK cat_ltm10877547.milestone3.mask_email
""")

result = spark.sql("DESCRIBE TABLE EXTENDED cat_ltm10877547.milestone3.ms3_s2_e05_customer_emails")
display(result)
df = spark.table("cat_ltm10877547.milestone3.ms3_s2_e05_customer_emails").orderBy("customer_id")
display(df)
```

Table

| | customer_id | customer_name | email | region |
|---|---|---|---|---|
| 1 | CE01 | Anita | a****@example.c... | NORTH |
| 2 | CE02 | Bala | b****@example.c... | SOUTH |
| 3 | CE03 | Charu | null | NORTH |
| 4 | CE04 | Deepak | d****@example.c... | WEST |
| 5 | CE05 | Esha | e****@example.c... | SOUTH |
| 6 | CE06 | Farhan | null | WEST |

2 dataframes: df, result

Table +

| | col_name | data_type |
|---|---|---|
| 1 | customer_id | string |
| 2 | customer_name | string |
| 3 | email | string |
| 4 | region | string |
| 5 | # Clustering Information | |
| 6 | # col_name | data_type |
| 7 | | |
| 8 | # Delta Statistics Columns | |
| 9 | Column Names | customer_id, customer_name, email, region |
| 10 | Column Selection Method | first-32 |
| 11 | | |
| 12 | # Detailed Table Information | |
| 13 | Catalog | cat_ltm10877547 |
| 14 | Database | milestone3 |
| 15 | | |

Evidence filename(s) / workspace reference:
___/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/finaleasy2_____
_____

### Task 4 - Verify protected query

Marks: 2 | Evidence: Code + Result

Required work: Query the managed table after mask activation and show masked count and NULL count.

Expected result: 4 MASKED emails; 2 NULL emails; 6 rows total.

Evaluator checks: Correct masked results and NULL preservation.

Screenshot instructions: Use MS3_S2_E02_T04_Code_A.png / Code_B.png for implementation, _Result_A.png / Result_B.png for outputs, and _UI_A.png / UI_B.png for configuration as applicable. Provide both implementation and observed result whenever the Evidence label includes both; use additional images for long code or separate metrics.

PASTE READABLE SCREENSHOT(S) FOR TASK 4 BELOW:

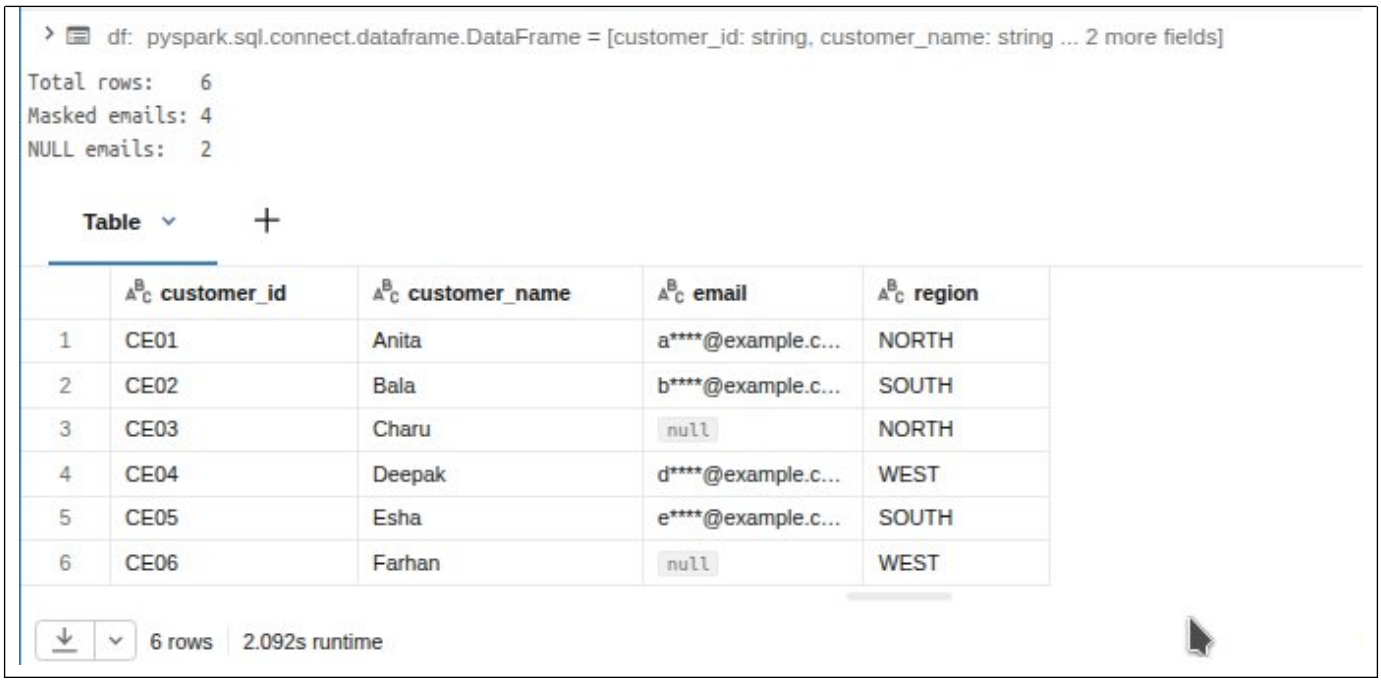
Screenshot holder (paste image here; duplicate this holder if you need multiple images)

```
df = spark.table("cat_ltm10877547.milestone3.ms3_s2_e05_customer_emails").orderBy
("customer_id")


masked_count = df.filter(col("email").rlike(r"^.\*{4}@")).count()
null_count = df.filter(col("email").isNull()).count()
total_count = df.count()

print(f"Total rows:    {total_count}")
print(f"Masked emails: {masked_count}")
print(f"NULL emails:   {null_count}")

display(df)
```




Evidence filename(s) / workspace reference: _____/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/finaleasy2_____

### Evaluation rubric

| Task | Criterion observable in submitted screenshots | Marks |
|---|---|---|
| 1 | Correct governed source setup | 2 |
| 2 | Correct managed table and NULL handling | 3 |
| 3 | Correct function and column-mask attachment | 3 |
| 4 | Correct masked results and NULL preservation | 2 |
| TOTAL | | 10 |

### Submission requirements

Submit the matching completed Word template with screenshots grouped under the corresponding tasks. Show code, settings, data outputs, and metrics according to the task-specific evidence list. Multiple images per task are allowed and encouraged when needed for readability or full rubric coverage. Do not submit invented results, hidden validation scripts, or a separate notebook as a replacement for screenshots.